The Pleistocene-Holocene Transition in Tagua-Tagua, central Chile

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Abstract

Paleohydrological reconstructions from the Chilean Altiplano document abrupt moisture fluctuations during the last millennia. Although the end of the mid Holocene aridity and the onset of more humid conditions between 6–4 ka has been identified in numerous regional marine and terrestrial sites, the timing of late Holocene dry and humid phases shows large regional variability. Laguna Miscanti and Laguna Miñiques (23° 43'S – 67° 46'W, 4140 m asl) are topographically closed, but connected by surface outflow from Miscanti. Sedimentological and geochemical indicators from two new cores show large facies changes, i.e. higher carbonate and evaporite deposition during more arid periods and increased organic productivity (both algal and macrophyte) during more humid phases. As in most Andean lakes located in volcanic settings, large 14C reservoir effects occur complicating 14C dating, so the age models include 210Pb and U/Th dating. In spite of dating uncertainties, both lakes show similar patterns during the last millennium. A humid phase in Laguna Miscanti prior to ca 1200 CE is coherent with rodent middens and geomorphological features indicative of a major pluvial/recharge event at lower altitudes (Atacama Desert) during the Medieval Climate Anomaly (ca 800 - 1300 CE). The LIA (1300 – 1850 CE) is characterized by several arid/humid cycles and the last century by a productivity increase. The hydrological changes observed during the last millennium illustrate the complex dynamics of recent climate evolution over the high altitude Andean plateau. Discrepancies between the northern and southern Altiplano records and with intermediate latitudes (Central Chile) records may reflect contrasting responses to external forcing (Westerlies versus South American Monsoon dynamics) along different climatic zones.

Key words: keywords

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1. Introduction

2. Study site and regional setting

3. Materials and methods

3.1. Sedimentary and geochemical analysis

In November 2019, we recovered a sedimentary sequence (~2.8 m) of the profile exposed on the western wall of the geoarchaeological excavation Tagua-Tagua 3 in ALLT (Figure 1). Sediment samples were obtained at 1-cm intervals and send at IPE-CSIC and IACT-CSIC laboratories in Spain for the geochemical and isotopic analyses. Subsamples were taken every 2 cm for analysed of total carbon (TC), total inorganic carbon (TIC), total organic carbon ($TOC = T - TIC$), total nitrogen (TN) and total sulfur (TS) in a LECO SCN furnace (IPE–CSIC). The grain-size frequency distributions between 0.02 um and 2000 um were measured in triplicate using a Malvern Mastersizer 2000 APA2000 grain-size analyzer with Hydro 2000G AWA2000. The sample sto determiner the grain–size were pre-treated with 10% H2O2 and 10% HCl to remove organic matter and carbonates, respectively. Robust End-member mixing analysis (EMMA) analisis was performed using the generic deterministic EMMA protocol (R package EMMAgeo, version: 0.9.8; Dietze et al 2012; Dietze and Dietze 2019) to infer the sediment sources, transport and depositional processes.

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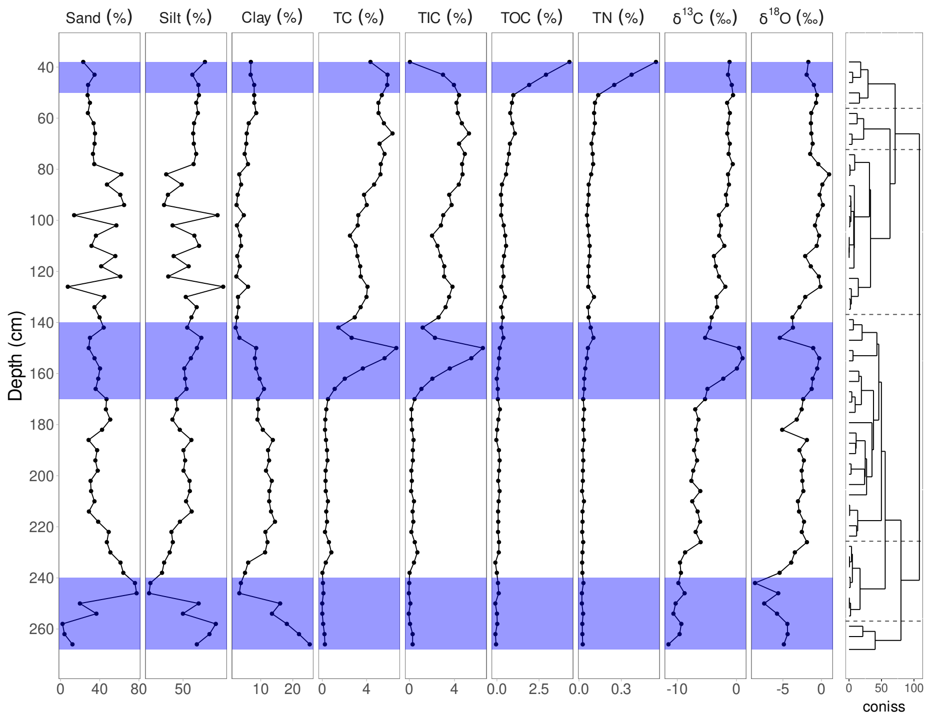
Elemental geochemical every 1 cm for total carbon (TC), total inorganic carbon (TIC), total organic carbon ($TOC = T - TIC$) and total sulfur (TS) in a LECO SC 144 DR furnace (IPE–CSIC). Total nitrogen (TN) was measured in a VARIO MAX CN elemental analyzer (IPE–CSIC). Grain–size distributions between 0.02 um and 2000 um were measured using a Malvern Mastersizer 2000 APA2000 grain-size analyzer with Hydro 2000G AWA2000. Sedimentary facies were described based on criteria formulated by (Schnurrenberger et al., 2003) and lacustrine units have been classified according to the sedimentary structure, MSCL data, elemental composition (TOC, TIC, TN, TS, TOC/TN, BioSi), XRF data and waxes composition.

3.2. Chronology

The radiocarbon sample on carbon and bulk sediment were obtained of the profile exposed sampled at 1 cm resolution (Table 1) . The radiocarbon ages were determined by AMS at DirectAMS laboratories, USA. Results are presented in units of percent modern carbon (pMC) and the uncalibrated radiocarbon age before present (BP).



4. Results

4.1. Cores correlations and sedimentary proxies

References

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